

# Dynamic-routing MVP findings — proxy-through-gluetun (Mullvad) e2e

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Operator-authorized (2026-07-02): reset podman + prove the full proxy-through-gluetun Mullvad e2e. This document records the honest results (§11.4.6) — what is proven and the real MVP dynamic-routing defects that block the full squid-integrated path.

## Proven live (rock-solid captured evidence)

- **podman/aardvark-dns is NOT broken** — throwaway inter-container DNS (10.89.2.2) and outbound DNS both resolve; the earlier "can't bind :53" fault has cleared. **No destructive podman system reset was done** — it would have destroyed the operator's 263 volumes (catalogizer/deploy/aosp/lava data — §9.2/§11.4.174).
- **squid-dynamic image rebuilt** (control-plane + squid multi-stage, config/squid context) → bakes http\_port :34128 (E's §11.4.108 staleness RESOLVED; source was already 34128).
- **Dynamic stack boots; gluetun WireGuard tunnel to Mullvad is UP** (exits observed Osaka / Denver / Malmö — public IP is a Mullvad exit each boot).
- **CORE proxy-through-gluetun egress PROVEN:**  
http\_proxy=http://proxy-gluetun:8888 → am.i.mullvad.net/json = mullvad\_exit\_ip:true (194.114.136.116, Osaka). Evidence qa-results/dynamic\_e2e\_20260702T175536Z/.
- **Squid dynamic-routing LOGIC proven:** un-routed target → fail-closed TCP\_DENIED/503 ERR\_TUNNEL\_DOWN (§11.4.68); with a route:<host:port> + up-status the acl-helper returns OK tag=<tunnel> and squid flips to TCP\_TUNNEL (allowed). The decision engine works.

# Open MVP dynamic-routing DEFECTS (the full squid-integrated path does NOT yet pass)

| #  | Defect                                                                                                                                                                                                                                                                                                     | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D1 | Compiler renders cache_peer gluetun-<profile> but the MVP ships one proxy-gluetun → peers unresolvable → all routed targets 503.                                                                                                                                                                           | <b>FIXED</b> 068dc78 (network aliases on the gluetun compose service).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| D2 | Route key is port-qualified: squid %>ha{Host} sends host:443 on CONNECT, so routes must be route:<host>:<port>.                                                                                                                                                                                            | Documented (compiler/doc note).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| D3 | <b>healthd↔gluetun health signal</b><br><b>BROKEN:</b> gluetun control API (/v1/publicip/ip, /v1/vpn/status) returns EMPTY; healthd's wg show <if> reads a WG interface in gluetun's netns (not healthd's) → every profile held falsely "down" → squid fail-closes even though the tunnel is genuinely UP. | <b>FIXED</b> 4a69225 (#79) + activated 2cd7f64. Took a THIRD path (neither doc option): healthd does a FRESH per-poll through-tunnel liveness PROBE via gluetun's built-in :8888 HTTP forward proxy — a real request egresses via the tunnel + returns THIS cycle (kill-switch blocks it when down ⇒ fail-closed). Additive to DecideHealth (§11.4.68-preserved, wg path intact), default-OFF until HEALTHD_TUNNEL_PROXY set. <b>LIVE-PROVEN:</b> the fixed healthd publishes state=up for the live Mullvad tunnel (qa-results/dynamic/d3_healthd_live_20260702T200259Z/ + d3_liveprobe_...195836Z/). |
| D4 | gluetun compose <b>healthcheck probes</b> /v1/openvpn/status for a WireGuard tunnel → gluetun labelled "unhealthy" though the WG tunnel is up.                                                                                                                                                             | <b>FIXED</b> 57cc857: healthcheck → VPN-agnostic /v1/vpn/status + a read-only gluetun control-auth config (gluetun v3.40 made control routes private/401) so healthd + the healthcheck stop 401ing. Live-validated (gluetun healthy; status 200; mutating routes 401) + §11.4.135 guard.                                                                                                                                                                                                                                                                                                              |
| D5 | Squid caches the gluetun-* peer                                                                                                                                                                                                                                                                            | <b>FIXED</b> 973ff59 (#81): a config/squid/wait-for-gluetun.sh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

| # | Defect                                                                                                                                                                                   | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <b>negative DNS</b> from its pre-alias startup; reconfigure did not clear it (peer stays dead). Needs squid to (re)resolve peers after gluetun is up (startup ordering / DNS lifecycle). | ENTRYPOINT wrapper (baked via Containerfile.dynamic) waits (bounded, fail-open-safe) for a gluetun-* peer alias to resolve, so squid's startup peer resolution is POSITIVE-cached, then execs the base entrypoint unchanged. Root cause confirmed by §11.4.150 research (squid resolves cache_peer names only at startup + caches NXDOMAIN with no reliable retry); podman-compose ignores depends_on: service_healthy so a container-level wait is the reliable mechanism. Independently reviewed GO (6/6); logic-validated standalone (resolve→exec, nxdomain→fail-open bounded). Engages on the next squid image rebuild. |

## Honest conclusion (§11.4.6)

The proxy-through-gluetun **egress capability** is proven live (gluetun's proxy egresses via Mullvad). **Rev 3 update (2026-07-02): ALL THREE deep blockers (D3, D4, D5) are now FIXED** (this session): D3 (the deepest — healthd false-down) by the fresh through-tunnel liveness probe (#79, LIVE-PROVEN healthd publishes state=up), D4 by the gluetun control-auth + VPN-agnostic healthcheck (live-validated), and D5 (squid peer negative-DNS) by the startup wait-for-gluetun ENTRYPOINT wrapper (#81, independently reviewed GO + logic-validated). Each fix's mechanism is proven live individually (D3 healthd=up; D4 gluetun healthy + auth-enforced; D5 the peer alias resolves on a fresh start). **Remaining to a passing FULL e2e: only the DEPLOY** — a ./start --dynamic that rebuilds the control-plane image (so the running proxy-healthd carries the #79 D3 binary — it's currently the old always-down build) and the squid image (so it carries the D5 entrypoint), then redeploys. With D3+D4+D5 all deployed + healthd=up: curl -x http://127.0.0.1:34128 <a routed host:port> egresses via the mullvad exit (mullvad\_exit\_ip=true), NOT the branded 503. Honest boundary (§11.4.6): the full squid-integrated e2e is **not yet claimed PASS** — the source fixes are complete + individually proven, but the composed e2e is proven only after the rebuild+redeploy. D1 fixed; D2 documented; **D3+D4+D5 FIXED**; full-e2e = the deploy step.